

Evaluation of Compressed Residential Energy Forecasting Models

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Abstract—Embedded energy gateways are increasingly being implemented for monitoring and control tasks in smart energy applications in buildings, cities, and localized electrical grids. These leverage state-of-the-art computational intelligence models for forecasting and anomaly detection. We present a study comparing various neural network model implementations for residential energy forecasting, both full-size models and compressed models using weight quantization and network pruning techniques. The evaluation is carried out on publicly available data consisting of 23 homes from the reference Pecan Street database. We analyse the optimal trade-off between accuracy, in terms of MSE quantitative metrics, and model complexity as reflected by network memory footprint for resource constrained embedded platforms. Using a pre-determined MSE threshold for residential energy forecasting, we achieve higher than 50% memory footprint reduction compared to the baseline scenario, using a combination of weight pruning and quantization, for both individual homes and global models that are trained across time series.

I. INTRODUCTION

Consumer-side energy management assumes robust models of demand which are trained on long-term power measurement time series and contextual information. Several types of models can be trained such as classical system identification, machine learning and deep learning models for short-term load forecasting (STLF) tasks. The main characteristics of these categories [1] are briefly summarised as follows:

- classical system identification: offers good performance while requiring expert knowledge for selecting the best model structure, such as ARMA, ARIMA, ARIMAX; the methods are highly specific to the process/phenomena being modelled;
- machine learning: it allows the incorporating of expert domain knowledge in the feature engineering stage; provides good performance and adequate explainability e.g. decision trees, largest margin classifiers;
- deep learning: offers state-of-the art performance with very large number of training examples but can be opaque in terms of explainability and require resource intensive training; built-in stepwise feature extraction.

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The predictive performance is highly dependant on both the quality and quantity of the training data with diverse datasets over longer periods of observation typically yielding the best results. State-of-the-art neural network structures exploit both long-term temporal and spatial dependencies to extract nonlinearities and perform fine grained discrimination. With regard to building energy consumption, daily, weekly and seasonal energy patterns are learned as well as variations from such patterns. Current challenges in electrical grid management imply ever decreasing time scales for measurement, inference and control in power systems. Edge inference for localized control of energy events [2] requires learning models that can output predictions under hard real-time constraints with low sampling rates. These typically run on resource constrained devices, which are closely coupled to and installed in physical proximity of the monitored equipment or process. The ability to achieve high quality predictions at the local level can be also used in anomaly detection and predictive maintenance for large scale equipment with economic and environmental impact.

Several methodologies have become available that allow the compression of full-scale neural network prediction and classification models to reduce their memory and computing footprint with bounded error metric degradation. The main categories identified and considered in this paper are: quantization - representing the model weight and bias terms on fewer bits in integer or binary formats and pruning - selecting the weights and neurons that contribute least to the output quality and discarding them. Suitable implementations as open source libraries and tools can be leveraged in a energy forecasting framework to select the lowest footprint algorithm that satisfies application performance metrics.

The main contribution of the work consists of a methodology to train, compress and evaluate energy forecasting models using state-of-the-art approaches and open source software instruments. This is accompanied by practical evaluation of the trade-off between model accuracy and model complexity in energy forecasting for dwellings. We compare a baseline fully connected feed forward neural network (FCNN) architecture trained on preprocessed features to various approaches for network compression. The results can be used for embedded deployment of the forecasting algorithms at the local level.